

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 8 - Formative Assessment

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Formative Assessment (FA-3)

Class: Class 8

Assessment Type: Formative Assessment (FA-3)

Total Marks: 30

Duration: 90 minutes

Topics Covered: SQL - Database Fundamentals

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

- a. Which SQL command is used to retrieve data from a database?
 - b. INSERT
 - c. SELECT
 - d. UPDATE
 - e. DELETE
- b. Which clause is used to filter rows in a SQL query?
 - c. GROUP BY
 - d. ORDER BY
 - e. WHERE
 - f. HAVING
- c. The SQL command `DELETE FROM students WHERE roll_no = 5;` will:
 - d. Delete the entire students table
 - e. Delete all rows from students
 - f. Delete only the row where `roll_no = 5`
 - g. Delete only the `roll_no` column
- d. Which data type is most suitable for storing a student's name in SQL?
 - e. INT
 - f. DATE
 - g. VARCHAR
 - h. BOOLEAN
- e. What does the `DROP TABLE` command do?
 - f. Removes all data but keeps the table structure
 - g. Removes the table structure and all data
 - h. Removes only duplicate rows
 - i. Removes all indexes

Section II: Short Answer (5 x 2 = 10 marks)

- Write the SQL command to create a table Books with columns: book_id (INT, Primary Key), title (VARCHAR 100), author (VARCHAR 50), price (DECIMAL), stock (INT).
- Differentiate between DELETE, TRUNCATE, and DROP commands.
- Write SQL queries to:
- Insert 3 records into the Books table
- Update the price of a book where book_id = 2
- Delete a book where stock = 0
- What is a **Primary Key**? What are its constraints?
- Write a SQL query to display all books from the Books table sorted by price in descending order, and only show books priced above ₹500.

Section III: Long Answer (5 x 1 = 5 marks)

- You are given a table Students with columns: roll_no, name, class, section, marks, grade. Write SQL queries for:
- Display students who scored more than 80 marks
- Count the number of students in each section
- Find the maximum, minimum, and average marks
- Display students sorted by marks in descending order
- Show only the top 5 students

Part B: Hands-on Practical Lab Task (5 marks)

Write a Python program that:

- Connects to an SQLite database (in-memory is acceptable)
- Creates a table Employees with columns: emp_id, name, department, salary, joining_date
- Inserts at least 8 records using user input or predefined data
- Performs 5 different SQL operations: SELECT with WHERE, ORDER BY, GROUP BY, aggregate functions, and UPDATE
- Displays results of each query in a formatted manner

Part C: Notebook Maintenance (5 marks)

- Neatness and organisation of notebook (2 marks)
- Completion of all assigned classwork and homework (2 marks)
- Proper headings, dates, and indexing (1 mark)